

DEPARTMENT OF ECONOMICS

UNIVERSITY OF CALABAR, CALABAR

SECOND SEMESTER EXAMINATION 2024/2025 SESSION

ECO 202: MICROECONOMIC THEORY.

TIME: 2 HRS

DATE: 23/06/2025

INSTRUCTIONS: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS ONLY.

1(a). Suppose the production functions of a cement manufacturing firm is estimated as:

$$Q = K^2 + K + 4L$$

where Q = quantity of cement in kilograms, K = Capital and L = Labour.

If the firm's total budget for the production of cement is \$102 million

Required:

(i) the optimal input combinations of K and L which will maximize the production of cement.

(ii) the maximum output obtained from the budget outlay.

1(b). Given that the short run production cost function for Lafarge Company Ltd is given thus:

$$TC = 150 + 50Q + 0.65Q^2$$
 where TC = Total cost and Q is the level of output.

Determine: (i) total fixed cost, (ii) total variable cost function, (iii) average variable cost function, and (iv) marginal cost function.

2a. Write short notes on the following:

i). Implicit, explicit and economic costs.

ii). Average Fixed cost and Average variable costs.

iii). The LRAC and LMC curves.

iv). The optimal plant size.

v). An expansion path.

2b. Give reasons for the U-shape of long run average cost curve.

3(a). With the aid of relevant graphs (diagram), demonstrate the existing relationship between Total product, Average product, and Marginal product and use same to explain the three (3) stages of production.

3(b). Explain the concept of Marginal Rate of Technical Substitution with relevant graphs.

3(c). Outline at least 5 properties of Isoquants.

4(a). Discuss four (4) sources where monopoly power can be derived from.

4(b). The relevant concept for describing long-run input-output relations is Returns to Scale, explain this in the light of the different types of Returns to Scale you know.

4(c). With the help of a graphical presentation, state and explain the term Isoquant and Isocost.

5a. The traditional economic analysis has four types of market. List and briefly explain any three.

5b. What are the features of a perfectly competitive market model?

5(c). Using the tools of Isoquant and Isocost, show graphically, the firm's output maximization decision subject to a cost constraint.